

**Some of you
will spend
THOUSANDS
to learn OLD
technologies**

**The smarter
ones will
grab every
opportunity
to master
the future**



**Read
LINUX For You**

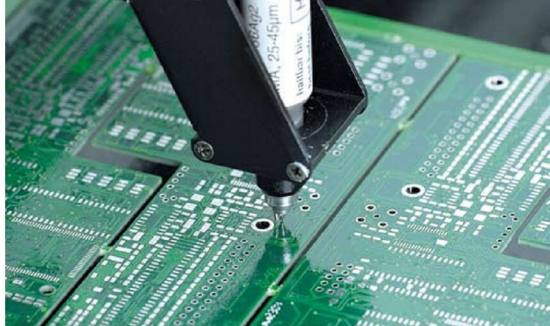
For more info, log on to:

www.linuxforu.com



ASIA'S FIRST
LINUX
MAGAZINE

LINUX For You
THE COMPLETE MAGAZINE ON OPEN SOURCE



the devices could be reduced to almost the size of the integrated circuit die itself.

The other package is in the form of a reduced-scale BGA, either in a partially or fully populated I/O condition.

"The increasing availability of the CSP significantly impacts the ability of product designers to design handheld products of a size and weight not previously possible," says Jain. The new CSP allows a higher density of components to be placed into an increasingly smaller portion of an existing PCB, or the PCB may be reduced with an accompanying reduction in product size, weight and cost.

Today, manufacturers are extensively using chip-scale packages along with different passive components to design devices such as palm-sized camcorders. This method helps to significantly minimise the time taken in transmission of signals since the components are placed closely. Consequently, this system produces a slew of faster circuits.

The smaller size of these components, along with their reduced weight, higher density and increased performance, has widely influenced the technologies and techniques used to assemble PCBs.

Assembly and component placement

Assembly techniques are becoming more and more precise, with demand for both smaller size and fine-pitched lead spacing. The entire gamut of placement machines and printers now addresses stringent requirements for speed and accuracy, notes Gurjal.

In spite of advancements in surface-mount technology, majority of circuit boards still contain a variety of through-hole components. Their electrical performance and mechanical requirements are satisfying enough, so these could not be completely thrown out of the scene.

Through-hole components such as connectors, electrolytic capacitors, power diodes, relays and transformers are odd-form in nature. (Odd-form stands for components with such a height, weight or shape that these cannot be placed automatically with standard pick-and-place machines.) Usually, odd-form components have been placed on circuit boards using discrete insertion machines, dedicated hard-tooled automation or manual assembly. Discrete insertion machines are best suited for applications with several odd-form components. Dedicated hard-tooled automation has been used to meet the needs of high-volume, low-

As technology is advancing, SMT components may have short pins or leads of various styles, flat contacts, a matrix of solder balls or termination on the body of components.